



Lecture 1~ Introduction

- Herbert Smith ~ 'satisficing'
- Post-optimality analysis → Addresses the underlying assumptions of the model and what if scenarios.
- Basic model terms:
 - The quantifiable decisions that need to be made: $x_1, x_2, x_3 \dots x_n$
 - Measure of performance - P described in objective function: e.g. $P = 3x_1 + 2x_2 + \dots + 7x_n$
 - Constraints on the decision variables: $2x_1 + 3x_2 \leq 7$
 - The goal would be to choose the values on $x_1 \dots x_n$ s.t. P is maximized.
- Sensitivity Analysis: Usually in post-optimality stage. Deciding which parameters cannot be changed without changing the optimal solution. Most important parameters.